Paper III
Gradient-Activated Phase Leakage and Vortex-Core Interiors in Coherence Gradient Gravity

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Abstract

We extend the nonlinear compact-object sector of Coherence Gradient Gravity (CGG) by introducing a gradient-activated approximate U(1) phase symmetry and a vortex nucleation criterion for rotating collapse interiors. Paper II established a smooth near-horizon basin ($\delta \geq 0$) in which strong-field deviations are bounded by log-envelope curvature while preserving exterior General Relativity (GR) behavior. The present work retains the GR event horizon as the causal boundary while allowing the interior to undergo a phase transition into metastable vortex filaments. Vortex nucleation occurs when rotational phase gradients exceed the healing-length threshold and phase-slip permission is activated by strong coherence gradients. Weak-field and exterior matching remain unchanged; observational deviations are restricted to dynamical ringdown coupling.

1. Continuity with Papers I–II

Paper II demonstrated that smooth nonlinear solutions ($\delta \geq 0$) form a GR-stable basin, with the proximity parameter ε controlled by log-envelope curvature near the Schwarzschild radius. Exterior solutions remain Schwarzschild/Kerr-like. Paper III does not modify the smooth branch or exterior metric structure; instead, it introduces an additional interior phase degree of freedom.

2. Field Architecture

We consider a complex coherence field $\psi = A e^{i\theta}$ with Mexican-hat potential $V(A) = (\lambda/4)(A^2 - \Psi_0^2)^2$. The coherence is defined as $C = |\psi|^2 = A^2$. The healing length is $\xi = c_s / \mathrm{sqrt}(2\lambda\Psi_0^2)$.

3. Gradient-Activated Approximate U(1) Symmetry

We introduce controlled explicit symmetry breaking activated by coherence gradients. Define $\chi = \ln(C + C_\varepsilon)$, where $C_\varepsilon > 0$ regularizes vortex cores. The breaking strength is defined as $\varepsilon_b(x) = \varepsilon_{b0} + \eta\, |\nabla\chi|^2$.

The modified Lagrangian includes $\Delta L_{break} = -\varepsilon_b(x)(\psi + \psi^*)$. The phase current $j^\mu = i(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*)$ satisfies $\partial_\mu j^\mu = 2\, \varepsilon_b(x)\, \mathrm{Im}(\psi)$, enabling controlled phase leakage.

4. Vortex Nucleation Criterion in Rotating Collapse

For axisymmetric interiors, let $\theta = m\varphi + \theta_r(r)$. Using cylindrical radius $\rho = r \sin(\vartheta)$, the phase-gradient magnitude is $|\nabla\theta|^2 \approx m^2/\rho^2 + (d\theta_r/dr)^2$.

A sufficient nucleation condition is $m^2/\rho^2 \geq \gamma_\xi/\xi^2$, which yields $\rho \leq \rho_* = (|m|/\sqrt{\gamma_\xi})\, \xi$. Vortex nucleation further requires $\varepsilon_{b0} + \eta|\nabla\chi|^2 \geq \varepsilon_*$.

5. Interior Vortex State

Within the nucleation region, amplitude $A \to 0$ along vortex filaments, forming decoherence cores. These structures redistribute phase energy without modifying the exterior Kerr-like solution.

6. Electromagnetic Propagation

Large phase gradients and coherence gradients inside the vortex region drive electromagnetic-like modes toward effective evanescence ($k^2 < 0$), producing interior attenuation without requiring geometric null trapping.

7. Ringdown Dynamics

Exterior quasi-normal modes remain Kerr-like. Interior vortex filaments introduce additional Kelvin/helical excitations that couple weakly to gravitational ringdown channels.

8. Falsification Criteria

The model fails if: (1) exterior multipoles deviate from Kerr beyond bounds; (2) Love-number suppression is not retained; (3) predicted echoes exceed observational limits; (4) vortex scales alter exterior behavior at $r > r_s$.

Conclusion

Paper III introduces gradient-activated phase leakage and vortex-core interiors while preserving the GR event horizon and exterior compact-object behavior. Interior strong-field

dynamics are governed by field-theoretic vortex physics without invoking geometric singularities.